



Preventing Readmissions

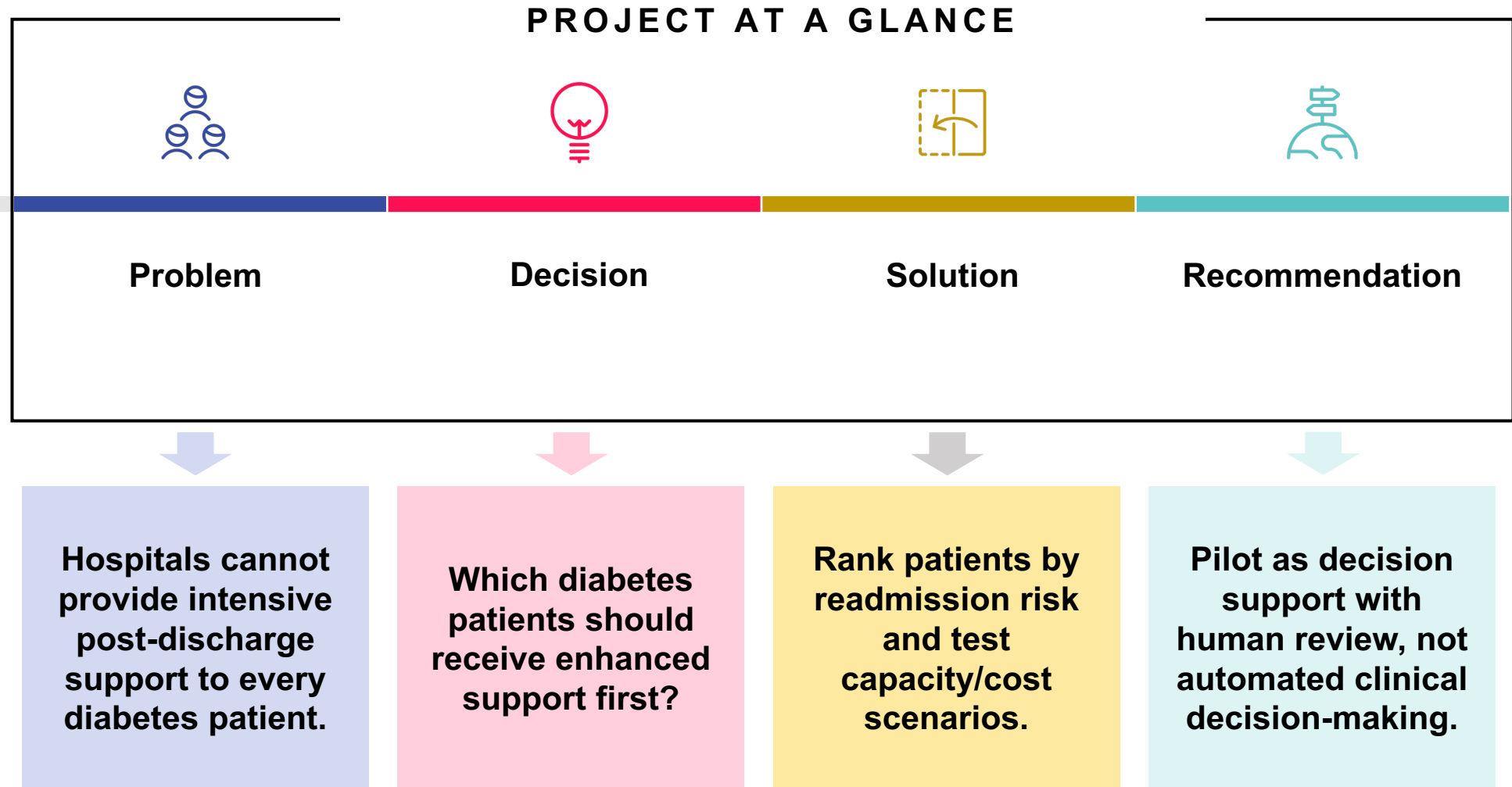
**Diabetes Transitional Care Resource
Allocation System**

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SMU[®]

Today's Agenda and Outcomes

From predicting readmission to deciding how to use limited care-management capacity



Business Problem & Stakeholder

- **Primary stakeholder:**

- Director of Care Management / Transitional Care Manager

- **Core constraint:**

- Limited staff, intervention slots, and budget

- **Decision:**

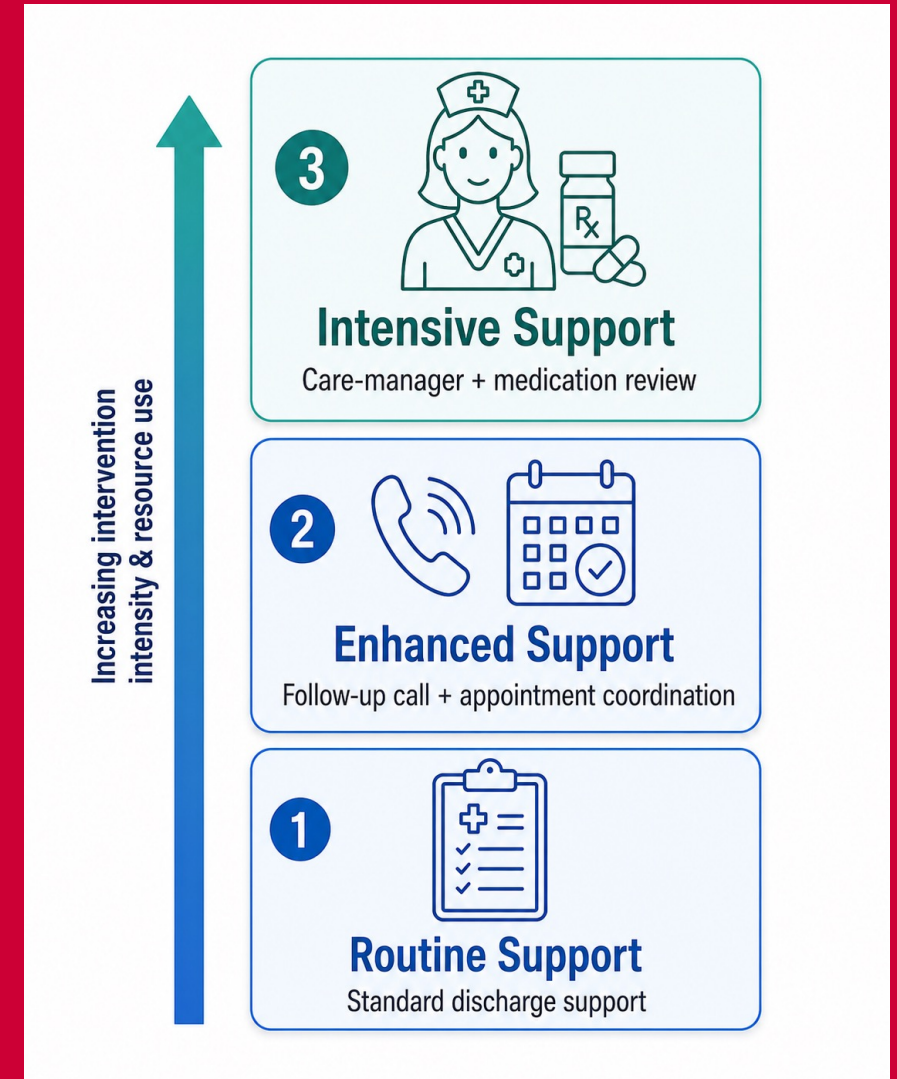
- Prioritize eligible diabetes discharges for the appropriate level of transitional-care support

- **KPI:**

- Recall@Capacity: How many actual readmissions are captured within the number of patients the team can serve?

- **Secondary Metrics:**

- Precision@Capacity · Lift · Expected Net Value



FACE Framework

Start from the discharge-support decision, not the dataset, then analyze, communicate, and execute against it.

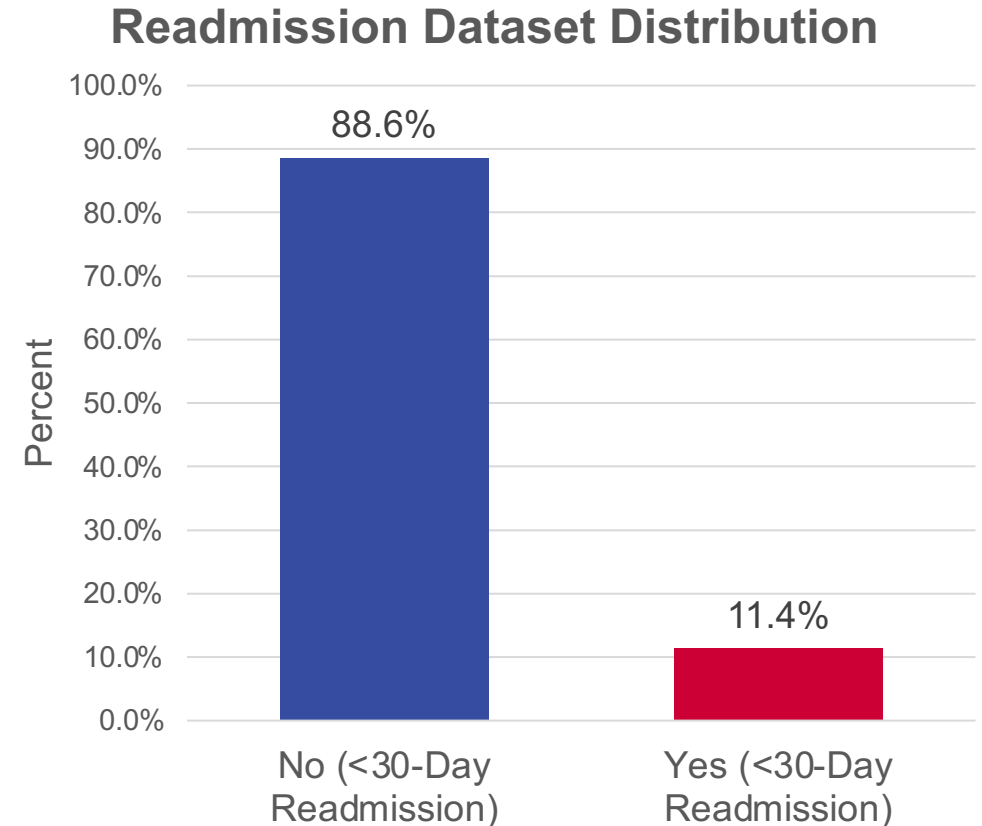
F Frame • Prioritize limited transitional-care resources for patients at highest readmission risk.	A Analyze • Clean and engineer data, compare models, evaluate ranking, build RAG, and estimate value.
C Communicate • Ranked worklist, risk tiers, capacity metrics, cost scenarios, and cited guidance.	E Execute • Pilot with care managers, validate locally, and monitor performance, fairness, and outcomes.

DELTA Plus: 7-Element Framework

- D** **Data** | Large but historical (1999 to 2008), with missing values and no intervention outcomes.
- E** **Enterprise** | Fits existing discharge-planning and care-management workflows.
- L** **Leadership** | Requires care management, nursing, pharmacy, IT, compliance, and finance support.
- T** **Targets** | Readmissions captured within capacity, targeting efficiency, and expected net value.
- T** **Technologies** | Python, scikit-learn, Gradio/Hugging Face, embeddings, and vector search.
- A** **Analysts** | Data science + clinical + care-management + finance expertise.
- A** **Analytical Techniques** | Classification, calibration, capacity metrics, RAG, and break-even analysis.

Data and Methodology

- **Dataset:** UCI Diabetes 130-US Hospitals dataset
 - 101,766 encounters
 - Diabetes inpatient encounters from 130 U.S. hospitals
- **Cleaned decision population:**
 - 99,340 eligible encounters
 - After excluding hospice and expired discharge dispositions
- **Target:**
 - Readmitted within 30 days = 1
 - Otherwise = 0
- 11.4% readmission rate
 - Imbalanced dataset
- **Feature engineering:**
 - Prior utilization, medication activity, length of stay, discharge/admission groups, A1C, ICD-9 chapters.



Model Selection: Gradient Boosting

Best fits the 500-patient capacity decision

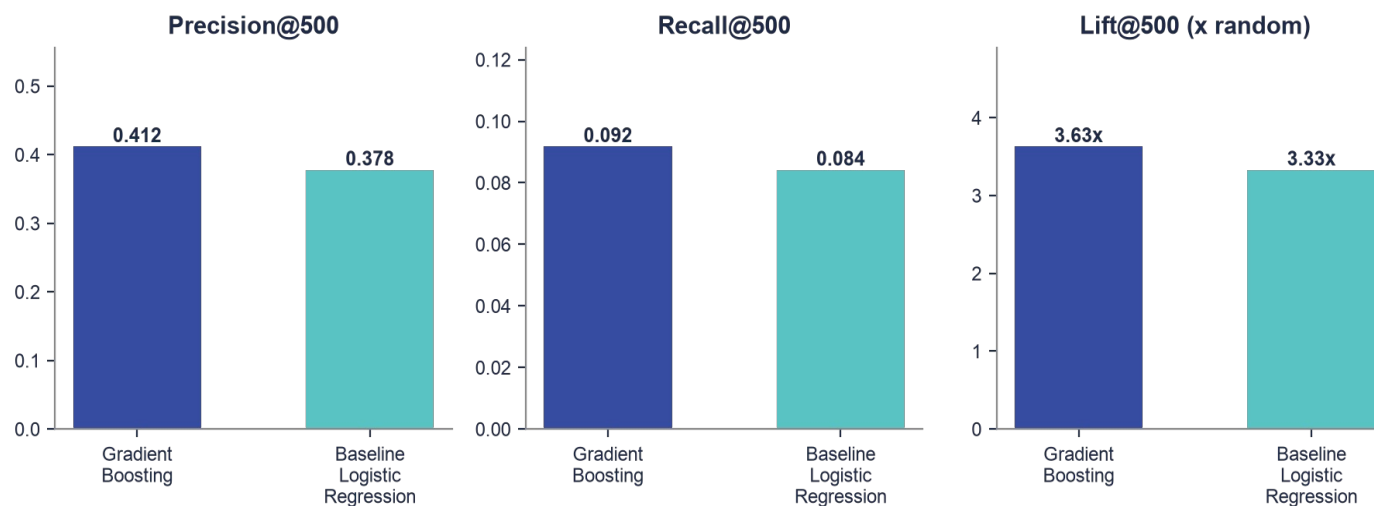
- **At 500 intervention slots:**

- 40.8% Precision: about 2 in 5 selected patients were readmitted
- 9.1% Recall: captured about 1 in 11 of all readmissions
- 3.6× Lift: identified readmissions 3.6× better than random selection
- 204 to 210 Readmissions Captured

- **Why Gradient Boosting?**

- Captured the most readmissions within the available 500 slots
- Best aligned with the goal of using limited care-management resources efficiently

Gradient Boosting vs. Baseline Logistic Regression at capacity 500



Risk Tiers Turn Scores Into Care Workflows

Low Risk

6.5% observed readmission rate
n = 9,887

Routine support

Medium Risk

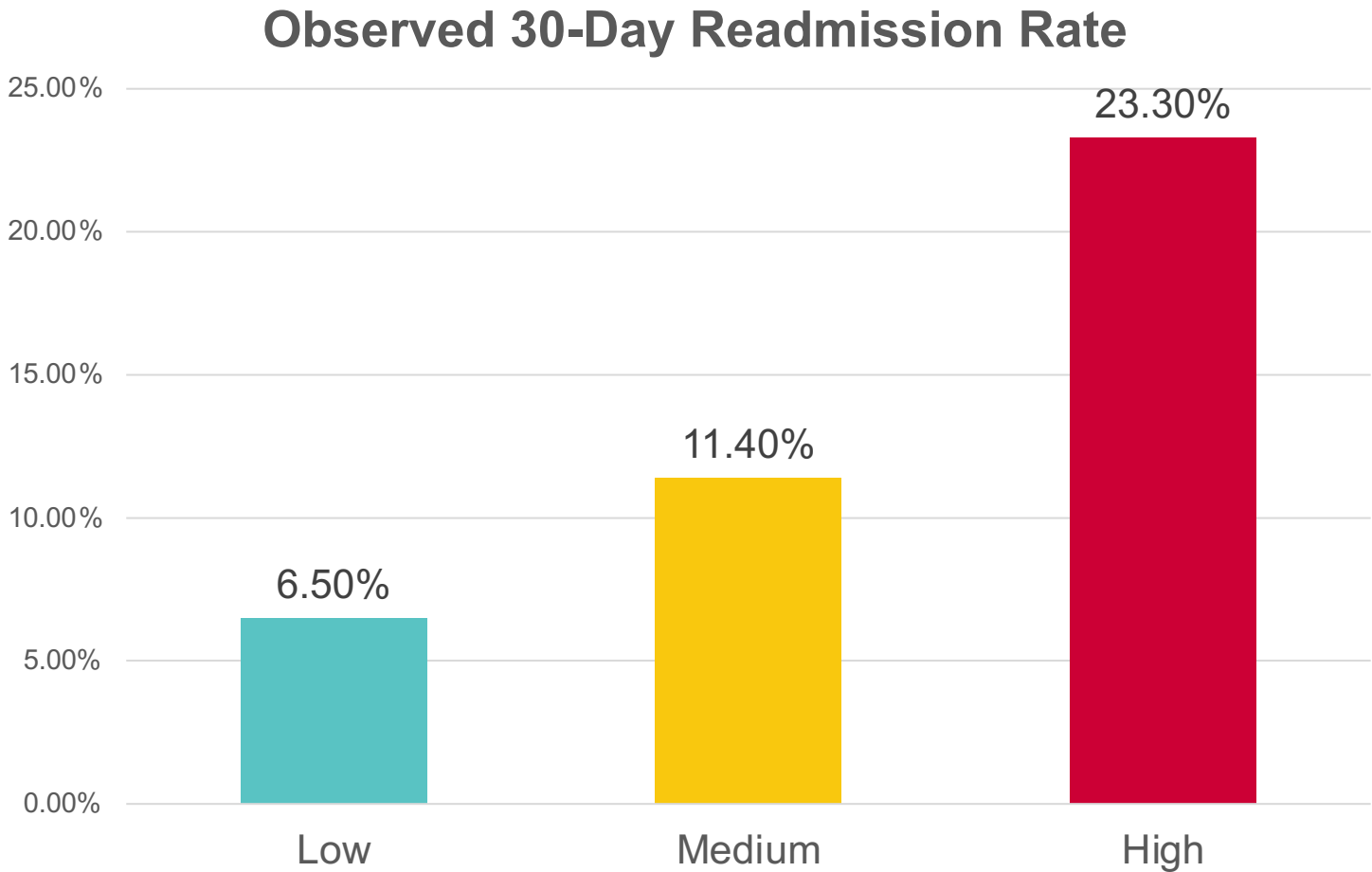
11.4% observed readmission rate
n = 5,934

Enhanced follow-up

High Risk

23.3% observed readmission rate
n = 3,952

Prioritized review for intensive support



High-risk patients readmitted at 3.6× the rate of low-risk patients.

Risk tier defines priority; available capacity determines how many patients are actually served.

RAG Connects Risk Scores to Care Guidance

- **Sources:** CMS HRRP + AHRQ Project RED/discharge resources
- **Retrieval:** MiniLM embeddings + FAISS vector search
- **Fallback:** TF-IDF retrieval if embeddings are unavailable
- **Topics:** medication reconciliation, follow-up, education, coordination

Model = who to prioritize

RAG = what evidence-based actions to consider

Ask: Care manager asks a discharge or care-transition question



Retrieve: Find relevant passages from approved CMS/AHRQ documents



Generate: Produce a concise answer with source and page citations



Apply: Care manager considers the guidance when planning follow-up

Fine-Tuning Pipeline: Business Evaluation

- 2,000 labeled prompts
 - Structured encounters were converted into text-based readmission examples.
- Binary Target
 - Readmitted within 30 days vs. not readmitted within 30 days
- Leakage Control
 - Evaluation remained patient-grouped.
- Common Scorecard
 - Compare models using PR-AUC, Recall@500, calibration, and expected net value.
- Goal
 - Determine whether a text-based classifier improves the actual care-management decision.



Structured Encounter → **Text Prompt** → **Readmission Classifier** → **Business Metrics**

Fine-Tuned Readmission Classifier: Model Comparison

Metrics	Gradient Boosting	Prompt Text Model
PR-AUC	0.241	0.147
Recall@500	9.10%	5.10%
Readmissions Captured	204	115
Expected Net Value	+\$113.7K	-\$14.3K

Gradient Boosting remained the better model for the resource-allocation decision.

Cost & Value: Scenario Assumptions

Turning cost and effectiveness assumptions into a net-value estimate

Assumptions



\$15,000

Assumed cost per readmission



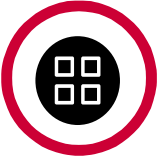
\$300

Assumed intervention cost per patient



10%

Assumed intervention effectiveness



500

Available intervention slots



What the Assumptions Produce



Expected Net Value

Avoided readmission cost –
intervention cost



Break-Even Question

How effective must the intervention
be, or how targeted must the worklist
be, for expected savings to exceed
cost?

*These are adjustable scenario assumptions, not
measured financial outcomes.*

Cost & Value: Results at 500 Patients

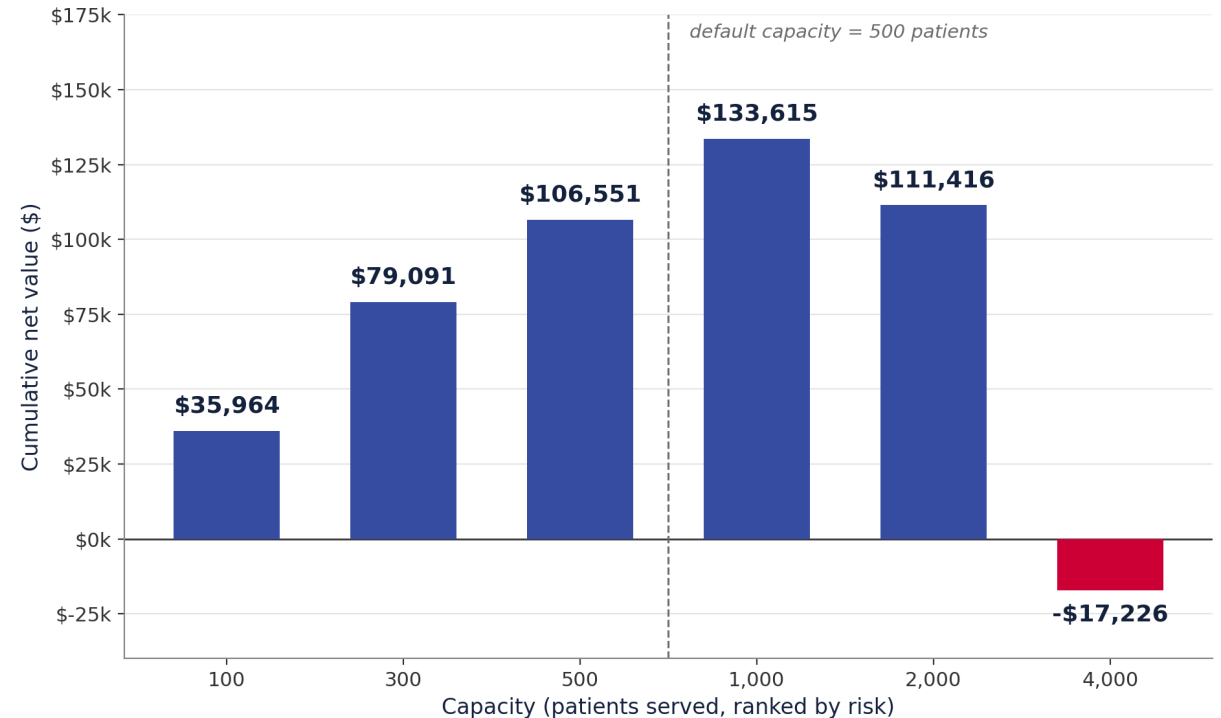
- **500 Patients Targeted**

- 175.8 expected readmissions without intervention
- 17.6 expected readmissions avoided
- \$263.7K gross avoided readmission cost
- \$150K intervention cost
- +\$113.7K estimated net value
- 5.7% break-even effectiveness

- **Key takeaway:**

- Targeting the top 500 patients creates positive expected value, but marginal value turns negative near 1,300 patients, so capacity should be optimized, not maximized.

Cumulative Net Value Stays Positive Well Past the Default 500-Patient Capacity



Product Demo

[Link to App](#)

Score

- Rank patients by 30-day readmission risk

Plan

- Adjust capacity, threshold, cost, and effectiveness

Guidance

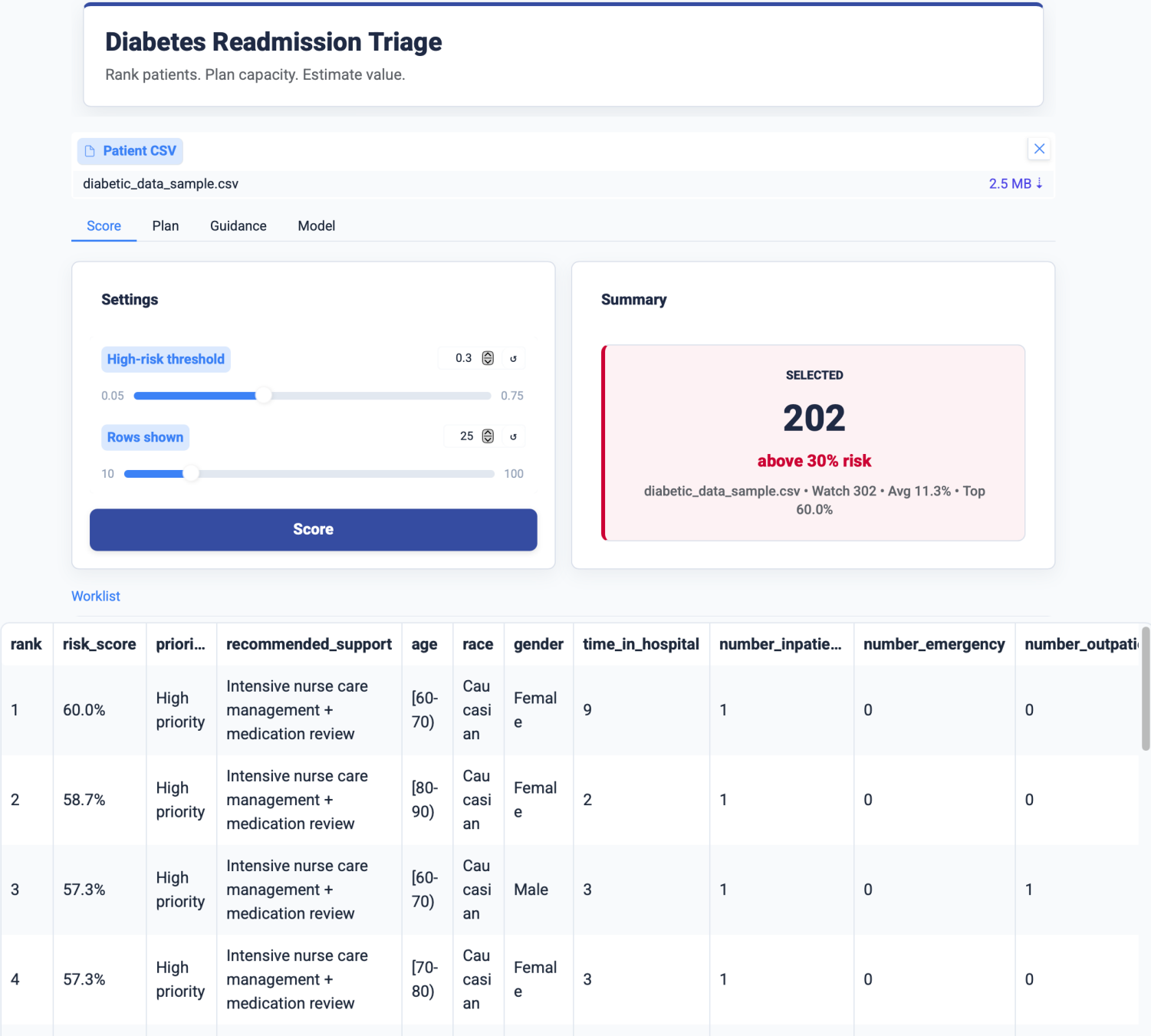
- Ask questions grounded in CMS/AHRQ documents

Model

- Review performance, limitations, and governance

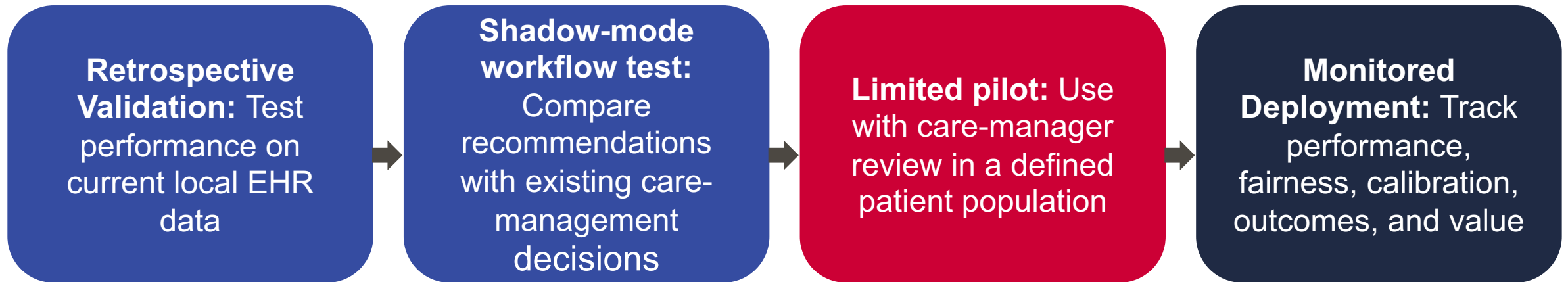
Care-Manager Workflow

- Upload or use a patient list
- Set available capacity
- Review prioritized patients and expected value
- Use guidance and governance information before action



Recommendation: Pilot as Decision Support

- Use risk ranking + capacity + net value to set the operating point.
- Use RAG for source-grounded care-transition guidance.
- Keep the final intervention decision with care managers.



The value is better prioritization under scarcity, not automated clinical decision-making.

Limitations, Risks & Next Steps

Limitations

- Historical UCI data from 1999 to 2008
- No observed transitional-care intervention outcomes
- Local validity has not been established
- Fairness and calibration require further validation
- RAG answer quality has not undergone formal evaluation
- Predictions show associations, not causal effects

Next Steps

- Validate on current local EHR data
- Monitor the fine-tuned prompt classifier against Gradient Boosting on future/local data.
- Build a formal RAG answer-quality test set
- Test the workflow with care managers
- Monitor ranking performance, fairness, calibration, and financial assumptions
- Complete privacy, governance, and EHR-integration review

Thank you!

Please reach out if you have
any questions



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Appendix



Appendix: Full Model Comparison Table

Metric	Gradient Boosting	Baseline Logistic Reg.	Balanced Random Forest	Balanced Logistic Reg.
Accuracy	0.8871	0.8876	0.6875	0.6632
Precision	0.6897	0.6400	0.1893	0.1815
Recall	0.0089	0.0214	0.5343	0.5606
F1	0.0176	0.0414	0.2796	0.2742
ROC-AUC	0.6756	0.6686	0.6706	0.6688
PR-AUC (avg precision)	0.2423	0.2239	0.2214	0.2233
Brier score	0.0951	0.0960	0.2086	0.2250
Precision@500	0.412	0.378	n/a	n/a
Recall@500	0.0918	0.0842	n/a	n/a
Lift@500 (x random)	3.6303	3.3307	n/a	n/a

19 Gradient Boosting was selected: best precision/lift at fixed capacity. Raw accuracy (~0.89) is misleading since only ~11% of patients readmit: always predicting "no" would already score ~89%, so ranking metrics (PR-AUC, precision/recall/lift at capacity) drove model selection instead.

Appendix: Confusion Matrix and Threshold Table

Default threshold (n=19,773)	Predicted not <30	Predicted <30
Actual not <30	17,524	5
Actual <30	2,235	9

Class	Precision	Recall	F1	Support
Not <30	0.89	1.00	0.94	17,529
<30	0.64	0.00	0.01	2,244

Overall accuracy at default threshold: 0.89.

Near-zero default-threshold recall is expected for a ranking model with a rare positive class.

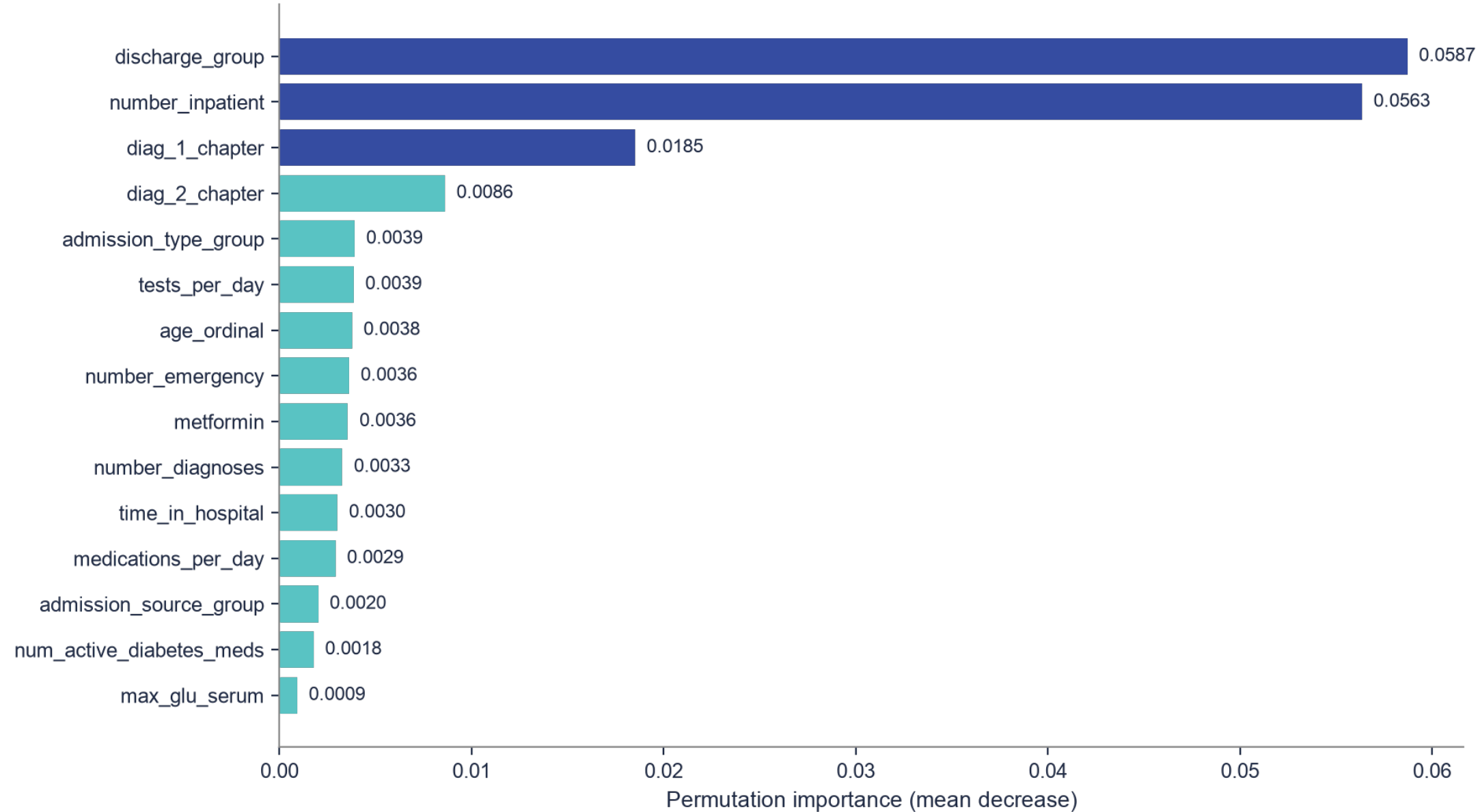
Threshold	Flagged	Flagged %	Precision	Recall	F1	Readmit. captured
0.05	17,649	89.26%	0.1231	0.9684	0.2185	2,173
0.10	10,098	51.07%	0.1606	0.7228	0.2628	1,622
0.15	3,557	17.99%	0.2407	0.3815	0.2951	856
0.20	1,133	5.73%	0.3345	0.1689	0.2245	379
0.25	626	3.17%	0.3882	0.1083	0.1693	243
0.30	359	1.82%	0.4596	0.0735	0.1268	~165
0.35	178	0.90%	0.5506	0.0437	0.0809	~98
0.40	80	0.40%	0.6500	0.0232	0.0448	~52
0.45	33	0.17%	0.6970	0.0102	0.0202	~23
0.50	14	0.07%	0.6429	0.0040	0.0080	~9

Appendix: Capacity Table by Patient Volume

Capacity	Precision@capacity	Recall@capacity	Lift over random	Readmissions captured
100	0.6400	0.0285	5.64x	64
250	0.5040	0.0561	4.44x	126
500	0.4200	0.0936	3.70x	210
1,000	0.3520	0.1569	3.10x	352
2,000	0.2835	0.2527	2.50x	567

As capacity grows, precision falls and recall rises: this tradeoff shapes staffing decisions. `model_summary.json` reports `recall@500` as 0.0918 with 204 captured (a very slightly different run than the notebook figures above); the deployed app uses the notebook capacity table.

Appendix: Feature Importance (Top 15)



Appendix: Fairness by Race, Gender, and Age

Race	n	Actual rate	Avg risk	Recall
Caucasian	14,813	11.72%	11.22%	0.52%
African American	3,697	10.79%	10.94%	0.00%
Unknown	447	7.16%	9.62%	0.00%
Hispanic	398	9.05%	10.27%	0.00%
Other	302	10.26%	9.87%	0.00%
Asian	116	8.62%	9.50%	0.00%

Gender	n	Actual rate	Avg risk	Recall
Female	10,726	11.45%	11.28%	0.65%
Male	9,047	11.23%	10.84%	0.10%

Key takeaway

Predicted risk scores track closely with actual observed rates across race, gender, and age; no group is wildly over- or under-scored. But recall at the default 0.5 classifier threshold is near zero for every group, because this is a ranking model, not a threshold classifier. Fairness should be evaluated on the ranked list / top-K, not on default-threshold recall. The [20-30) group has the highest actual rate but a small sample size (n=369).

Age group	n	Actual rate
[10-20)	137	5.84%
[20-30)	369	16.26%
[30-40)	757	11.49%
[40-50)	1,909	9.59%
[50-60)	3,467	10.87%
[60-70)	4,356	11.57%
[70-80)	4,921	11.30%
[80-90)	3,284	12.15%
[90-100)	547	12.43%

Appendix: Cost / Value Formulas

Quantity	Formula
Expected readmissions without intervention	capacity x baseline_readmission_rate_among_selected
Expected readmissions avoided	expected_readmissions_without_intervention x intervention_effectiveness
Gross avoided readmission cost	expected_readmissions_avoided x readmission_cost
Total intervention cost	capacity x intervention_cost
Net value	gross_avoided_readmission_cost - total_intervention_cost
Break-even effectiveness	total_intervention_cost / (expected_readmissions_without_intervention x readmission_cost)

Default assumptions: readmission cost \$15,000; intervention cost \$1,200; intervention effectiveness 10%; capacity 500; threshold 0.0. These are scenario inputs the app lets a care-management leader adjust; they are not measured financial outcomes.

Appendix: RAG Corpus and Example Questions

Corpus (pdfs/): HRRP_Formula.pdf (CMS HRRP payment reduction formula) | A Reengineered Hospital Discharge Program to Decrease Rehospitalization.pdf (AHRQ Project RED trial) | dischargeprep.pdf (RED discharge preparation workbook) | goinghomeguide.pdf (AHRQ going-home guide)

Requirement	Status	Notebook / app connection
Document collection	Done: 4 approved CMS HRRP / AHRQ RED PDFs in pdfs/	Guidance tab corpus
Document loading & preprocessing	Done: pypdf text extraction, cleaned and paginated	RAG preprocessing script
Chunking strategy	Done: ~950-character chunks, 160-character overlap, sentence-aware breaks	Retriever quality
Embeddings	Done: sentence-transformers/all-MiniLM-L6-v2 dense embeddings	Retriever index
Vector search / retrieval	Done: FAISS vector search plus topic keyword boosting (TF-IDF fallback if embeddings unavailable)	Question-answer workflow
Prompt design	Done: system prompt scopes the assistant to grounded, cited, non-clinical guidance	Business-facing answer behavior
Generated response	Done (optional): OpenAI gpt-4o-mini answer when a user key is supplied, otherwise an extractive summary	Stakeholder output
Source grounding	Done: source title, page, and snippet shown with every answer	Trust and auditability
Evaluation & failure modes	Partial: manual test questions only, documented fallback behavior, no formal answer-quality rubric yet	Limitations and governance

- Example care-transition questions:**
- What should a care manager confirm before a high-risk diabetes patient is discharged home?
 - What does the AHRQ Re-Engineered Discharge (RED) approach recommend for medication reconciliation?
 - How does CMS calculate the HRRP readmission payment penalty?
 - What follow-up appointment timing does the AHRQ discharge guidance recommend for high-risk patients?

Appendix: Fine-Tuning Example Record

2,000 fine-tuning examples in artifacts/fine_tuning_readmission_examples.jsonl. Real example record:

text: "Classify whether this diabetes inpatient encounter is readmitted within 30 days. Age: [60-70); race: Caucasian; gender: Male; time in hospital: 4; prior inpatient visits: 0; prior emergency visits: 0; prior outpatient visits: 0; number of medications: 13; number of diagnoses: 7; A1C result: >8; insulin: 0; discharge group: Home; admission source: Emergency room."

label: "not_readmitted_within_30_days"

Reusable technical pattern:

1. Base model: Qwen 2.5 1.5B
2. Efficient fine-tuning: Unsloth LoRA adapters
3. Forced-choice label scoring (constrain generation to the label set instead of free text)
4. Save and reload the fine-tuned model via the Hugging Face Hub
5. Serve inference through a Gradio interface